

Ammara Noor

AI / ML Engineer - Computer Vision, LLM & RAG Systems, Production ML

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Available immediately | Open to Lahore / Islamabad / Karachi / Remote

SUMMARY

AI/ML engineer who takes systems from data collection through to production deployment. Took a fabric inspection line from fully manual to 95%-accurate automated defect detection with YOLOv8 and PatchCore, collecting and labelling the ~1,600-image dataset on-site. Shipped production RAG pipelines into live client applications, and reproduced PatchCore from the original paper to 99.56% AUROC on MVTec AD. 1st place, ICAT National Robotics Competition 2025.

TECHNICAL SKILLS

Computer Vision: Object detection (YOLOv8), image classification, segmentation, anomaly detection (PatchCore), OCR, object tracking, real-time video analytics, image processing, dataset collection and labelling, Roboflow

Machine Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face, Ultralytics, OpenCV, NumPy, Pandas, training, fine-tuning, transfer learning, evaluation (AUROC, mAP, F1, precision/recall), MLflow experiment tracking, threshold calibration

LLM & Generative AI: Retrieval-augmented generation (RAG), embeddings, vector search, chunking and retrieval strategy, prompt engineering, LLM API integration, structured outputs, document processing

MLOps & Deployment: FastAPI, Flask, REST API design, Docker, Linux, Git, ONNX export, latency profiling, inference optimisation, drift monitoring, on-site installation, live camera and hardware integration

Backend & Frontend: Node.js, Express.js, MySQL, SQLite, role-based authentication, React.js, Next.js, operator dashboards, live-updating interfaces

Languages: Python, JavaScript (ES6+), SQL, C/C++ | English (professional working proficiency), Urdu (native)

EXPERIENCE

AI / Computer Vision Engineer (Part-Time) - Robionix Technologies, Lahore, Pakistan

Feb 2026 - Jun 2026

Built and deployed MagicQC, a fabric defect detection system, from scratch to a live production line at 95% accuracy on a held-out test set.

Collected and labelled ~1,600 images on-site at a textile mill; trained YOLOv8 for three defect classes and added PatchCore anomaly detection to catch defect types absent from the training set.

Shipped the full pipeline solo - camera capture, OpenCV preprocessing, dual-model inference, Flask API, React operator dashboard, Dockerised deployment - then installed it on the line, ran operator rollout, and re-tuned thresholds against real production conditions to cut false positives.

Exhibited at the 32nd Textile Asia Expo - Nishat Mills, Gul Ahmed and Sapphire requested evaluations; presented the technology and business case at a national industry open house.

Full-Stack AI Developer - Evolvian Softwares, Lahore, Pakistan

Aug 2025 - Jan 2026

Shipped production RAG pipelines end to end - document ingestion, embeddings, vector retrieval, LLM generation - into live client applications.

Rebuilt chunking and retrieval strategy after diagnosing why standard metrics did not predict real failures, raising output consistency across messy client documents.

Collapsed a multi-step manual review process into one automated pipeline by chaining LLM text processing, OCR and YOLOv8 detection.

Built the REST APIs and React dashboards with role-based auth and live updates; owned features from client requirement to deployed release.

AI & UAV Engineering Intern - National Development Complex (NESCOM), Islamabad, Pakistan

Jul 2025 - Sep 2025

Built a UAV surveillance system performing person tracking and vehicle detection on live aerial video feeds.

Delivered a smart parking dashboard rendering live detections and slot occupancy from the raw detection stream.

PROJECTS

Industrial Anomaly Detection - MLOps Pipeline - PyTorch, PatchCore, MLflow, FastAPI, Docker, ONNX

github.com/ammaraan620-de/industrial-anomaly-detection-mlops

Implemented PatchCore from the original paper - frozen WideResNet-50 features, patch memory bank, 1% greedy k-center coreset, trained on defect-free images only.

99.56% AUROC on MVTec AD carpet: 87 of 89 defects caught, 26 of 28 good pieces passed, precision/recall/F1 all 0.9775 at the 95th-percentile operating threshold.

Served through FastAPI with MLflow-tracked runs, Docker Compose and a drift signal; profiled 535 ms/image on CPU against the 33 ms a 30 FPS line needs and documented the ONNX/INT8 path.

EDUCATION

BS Computer Engineering - National University of Technology (NUTECH), Islamabad

2022 - 2026

CGPA 3.44 / 4.00 | Coursework: machine learning, deep learning, computer vision, databases, data structures and algorithms, OOP.

ACHIEVEMENTS & LEADERSHIP

1st Place, ICAT National Robotics Competition 2025 | Runner-Up, NAIS 2025 National AI Seminar | Millennium Fellow, United Nations Academic Impact Programme.

Chair, IEEE Women in Engineering, NUTECH Student Branch | Chief Coordinator, NUTECH Student Council.

Ambassador, 6th International Student Convention & Expo 2026, Islamabad | Exhibitor, 32nd Textile Asia Expo.